

Benchmark - Security Controls Analysis

Grand Canyon University

ITT-430 IT Security Driven Systems Administration

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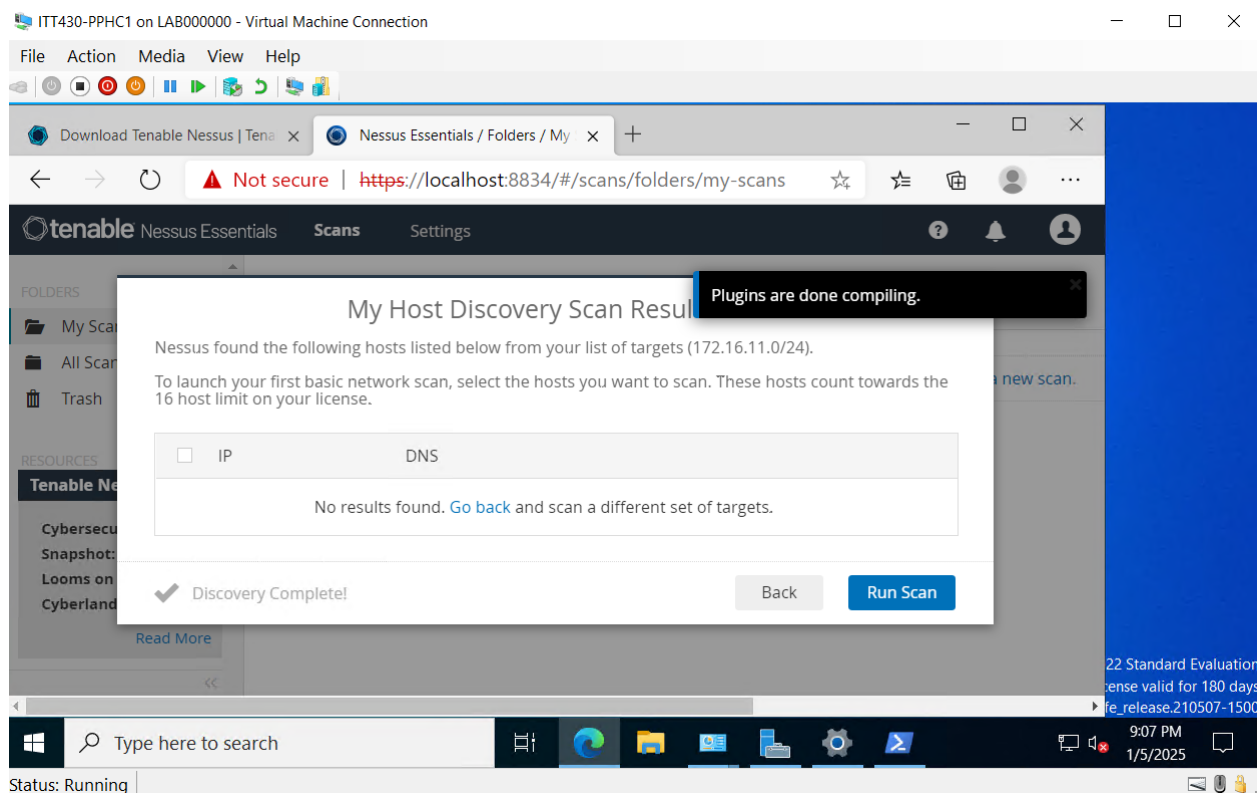
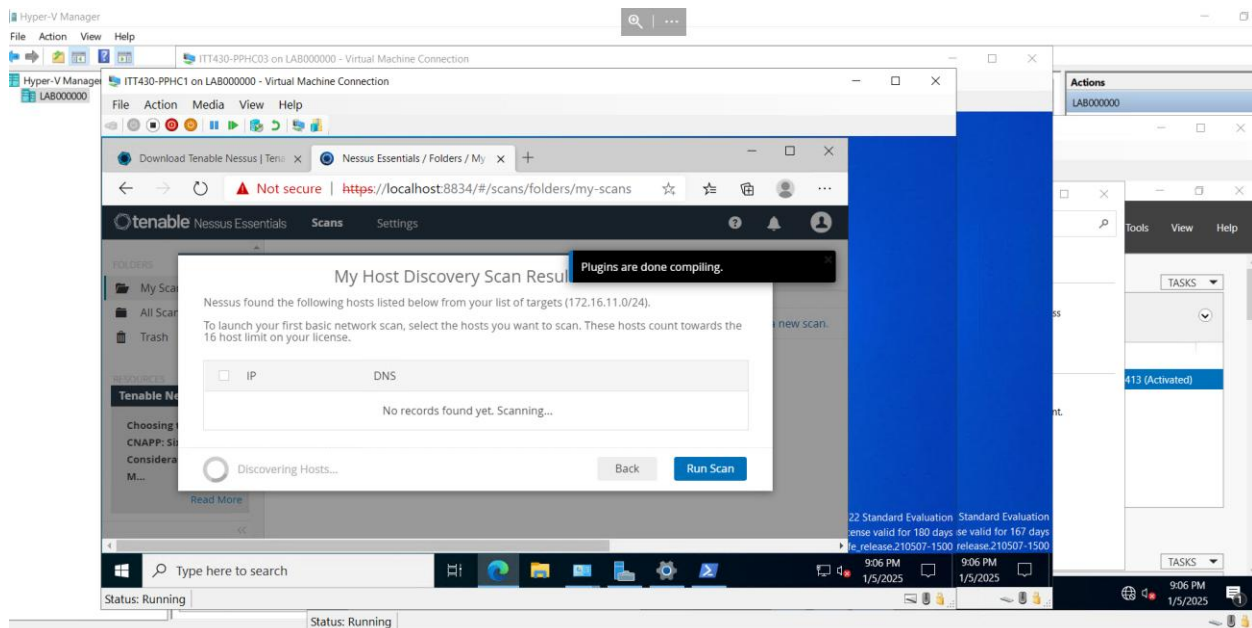
January 10, 2024

Benchmark - Security Controls Analysis

Nessus Essentials was downloaded to perform a network vulnerability scan through the PPHC DOnain Network prior to updates. Since each server has updates being installed automatically, it is no surprise that prior to the intentional update that the Nessus scan revealed 0 high risk vulnerabilities but revealed basic information about the servers along with a few vulnerabilities that produced no risk but could be fixed. Servers 2 and 3 successfully updated while Server 1 updated caused it to break and reverse the update and clean up in order for it to even connect. As seen in previous screenshots. After the update, the scans revealed similar results to the first as there were no risk vulnerabilities but revealed tasks that can be completed to better the defense such as closing a SYN half open port scanner. On Server 2 there were 15 vulnerabilities found each with 0 risk and were labeled as information, Server 3 found 14 vulnerabilities each with 0 risk and labeled as information.

To have proactive security procedures in response to the findings, I would recommend IDS software and ensuring that firewalls protection protocols continue to be automatically updated. Firewalls allow for unauthorized access to the network by using packet filtering, examining data packets and ensuring that they do not contain malicious data while matching criteria that defines a threat. Next I would continue to implement a strong password policy across all servers to ensure that unauthenticated users will have more difficulty gaining user access. In addition to the IDS software, it is recommended that antivirus and malware software have continuous monitoring, threat identification, regular scans for viruses and prompts if detected. As already implemented, having network access control protocols implemented is necessary to reduce the potential damage a unauthorized user/malicious attacker can have during a breach.

For example ensuring that all users provisions follow a least privilege protocol which ensures that users are only given restricted access to specific capabilities that they require to perform their job efficiently. Next, ensure that each server is running their updates regularly and automatically; windows servers install updates and patches for security. Next a VPN, Virtual Private Network, allows servers to encrypt data and mask IP addresses which allows for secure connections between each endpoint and the remote server which is essential for each of the servers not only to connect to the DC server but also to the internet. Intrusion Prevention Systems are a great way to be proactive in security measures as they operate in the network right behind the firewall which inspects packets. If the system detects an anomaly then it sends the IT admin a notification and works to deactivate the threat. Most importantly, PPHC will include a backup cycle as laid out in the incident recovery plan within the security policy. This provides a recovery strategy for data and the systems to allow for business operations in the case of a data breach, employee error or another incident.



ITT430-PPHC1 on LAB000000 - Virtual Machine Connection

File Action Media View Help

Download Tenable Nessus | Tena x Nessus Essentials / Folders / My x +

Not secure | https://localhost:8834/#/scans/folders/my-scans

tenable Nessus Essentials Scans Settings

My Host Discovery Scan Results

Plugins are done compiling.

Nessus found the following hosts listed below from your list of targets (CORP.ITT430PPHC.COM).

To launch your first basic network scan, select the hosts you want to scan. These hosts count towards the 16 host limit on your license.

☐	IP	DNS
No results found. Go back and scan a different set of targets.		

Discovery Complete! Back Run Scan

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tenable Nessus Essentials Scans Settings

My Scans

Import New Folder New Scan

Search Scans 5 Scans

☐	Name	Scan Type	Schedule	Last Scanned		
☐	My Basic Network Scan	Vulnerability	On Demand	Today at 9:18 PM		■
☐	HOST 3 PREUPDATE	Vulnerability	On Demand	Today at 9:18 PM		■

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Not secure | <https://localhost:8834/#/scans/reports/14/hosts>

tenable Nessus Essentials Scans Settings ? gcuLPres

FOLDERS

- My Scans 3
- All Scans
- Trash

RESOURCES

Tenable News

Progress WhatsUp
Gold NmAPI.exe
Registry Overwrite...

Read More

My Basic Network Scan

[Back to My Scans](#)

Hosts 1 Vulnerabilities 5 History 1

Filter Search Hosts 1 Host

Host	Vulnerabilities	%
172.16.111.3	11	7%

Hyper-V Manager

Scan Details

Policy: Basic Network Scan
Status: Running
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 9:13 PM

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Not secure | <https://localhost:8834/#/scans/reports/17/hosts>

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FOLDERS

- My Scans 3
- All Scans
- Trash

RESOURCES

Tenable News

SQL Injection in
WordPress Project
Manager Plugin

Read More

HOST 3 PREUPDATE

[Back to My Scans](#)

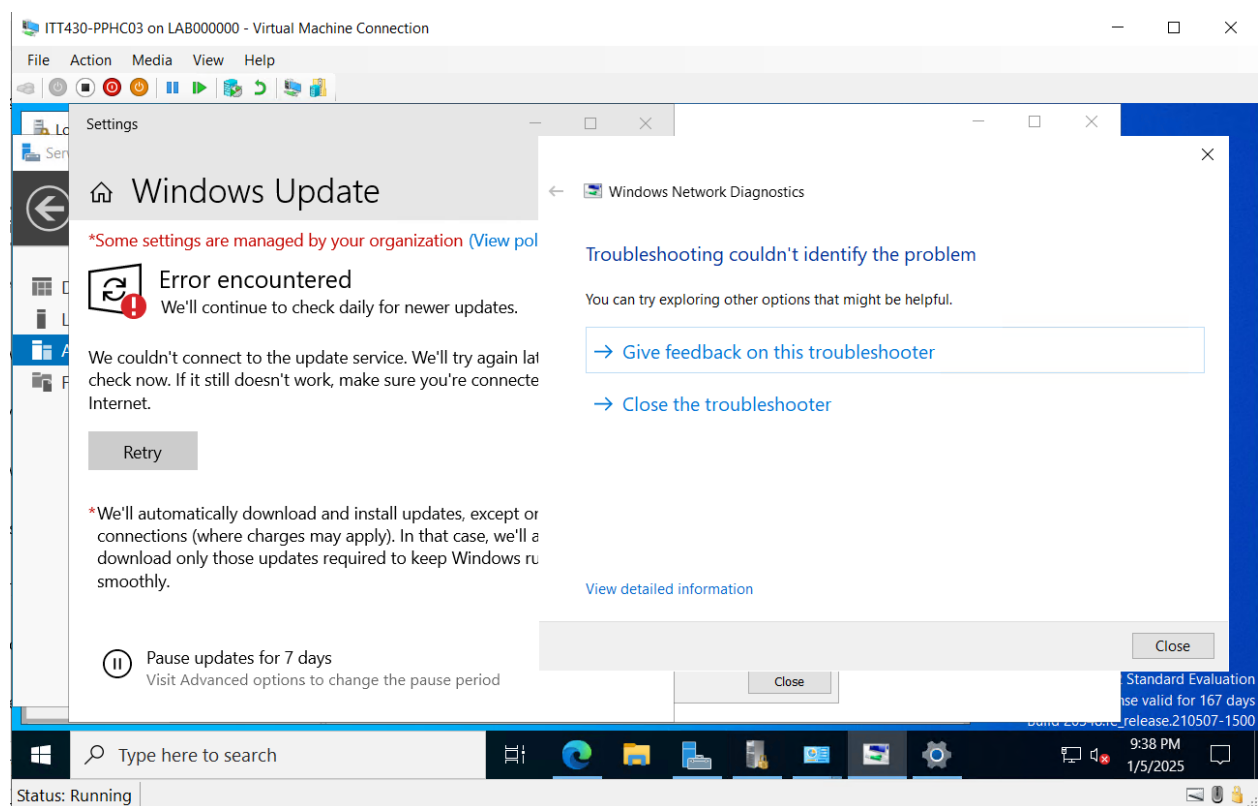
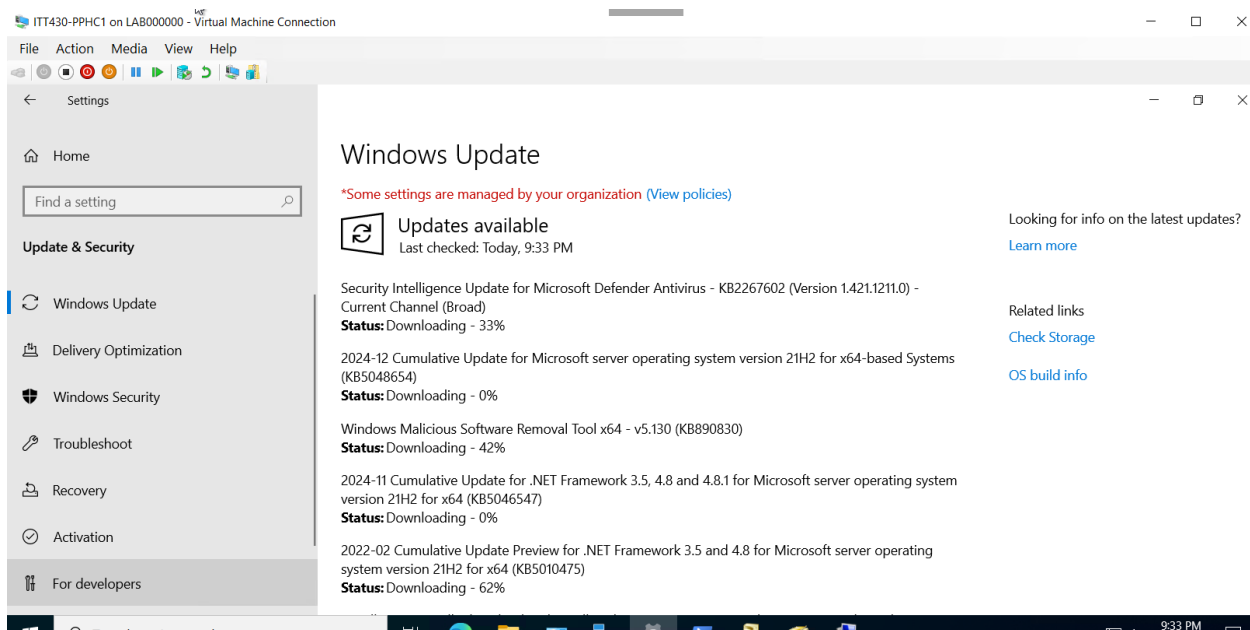
Hosts 1 Vulnerabilities 2 History 1

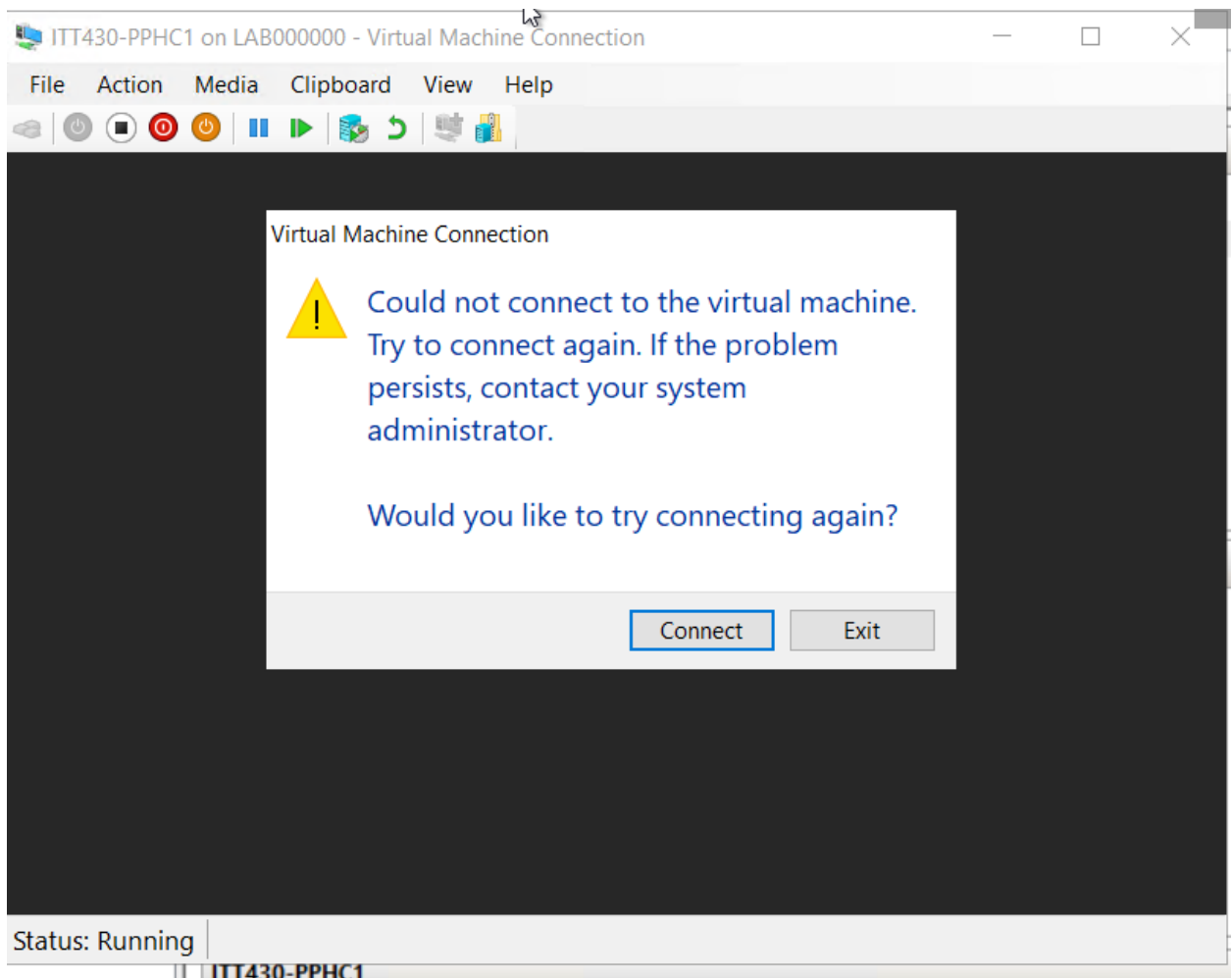
Filter Search Hosts 1 Host

Host	Vulnerabilities	%
itt430-pphc3	2	0%

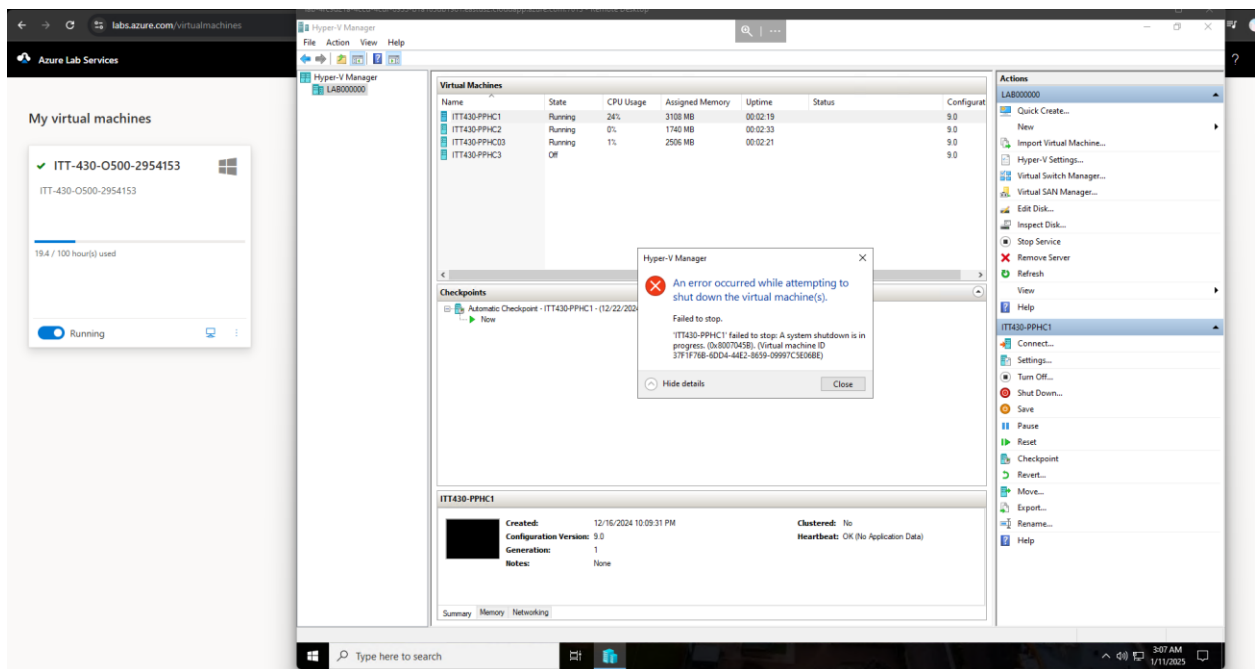
Scan Details

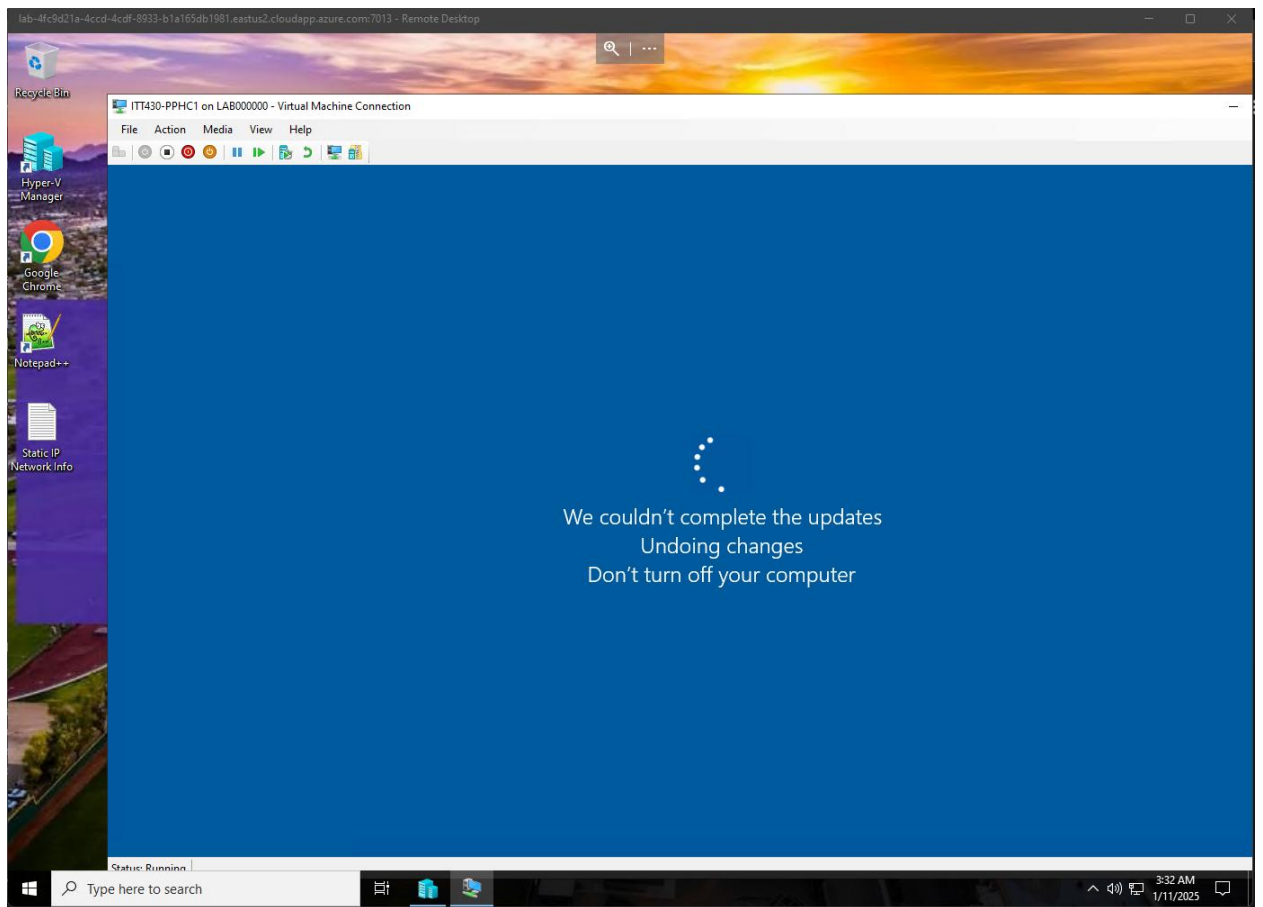
Policy: Basic Network Scan
Status: Running
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 9:17 PM

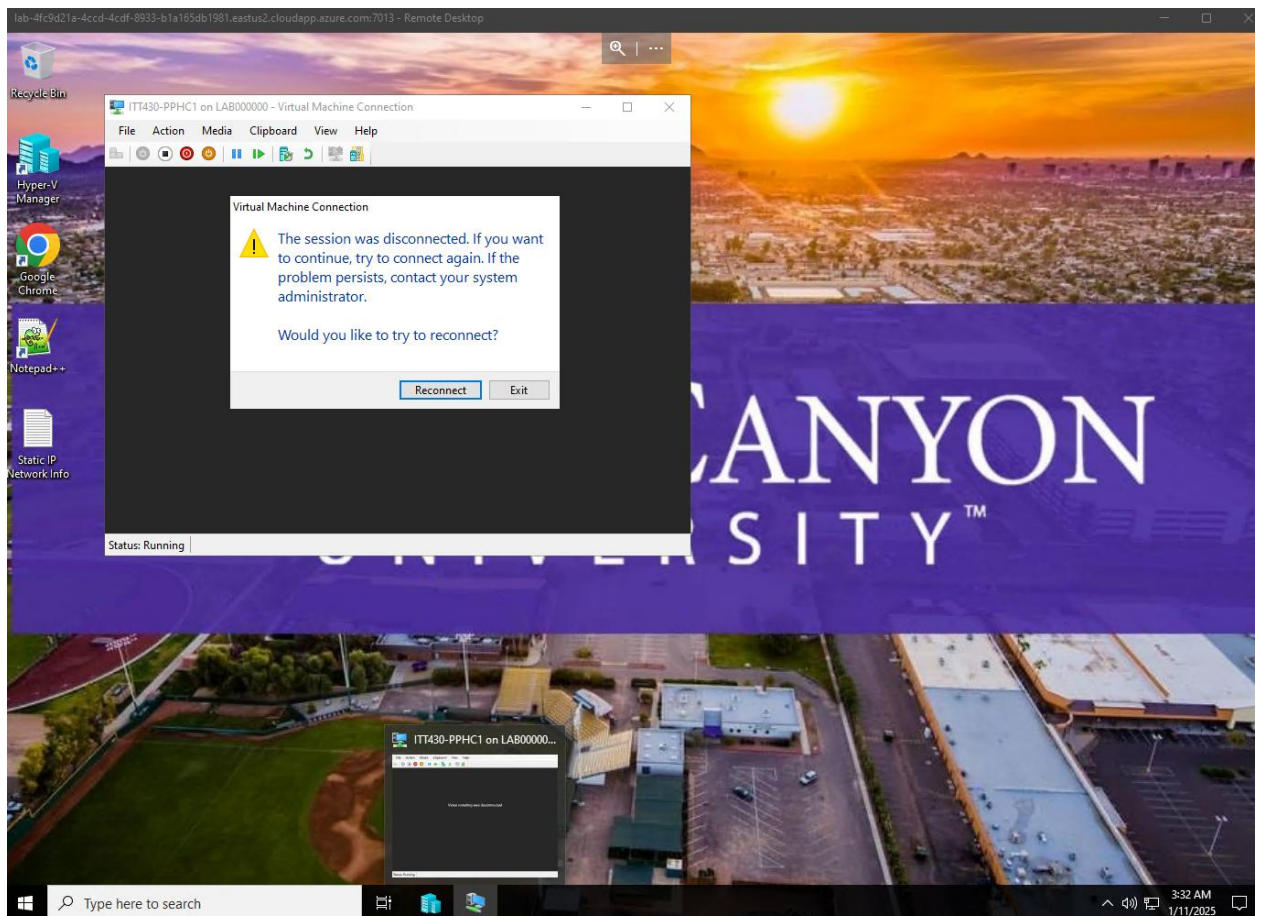


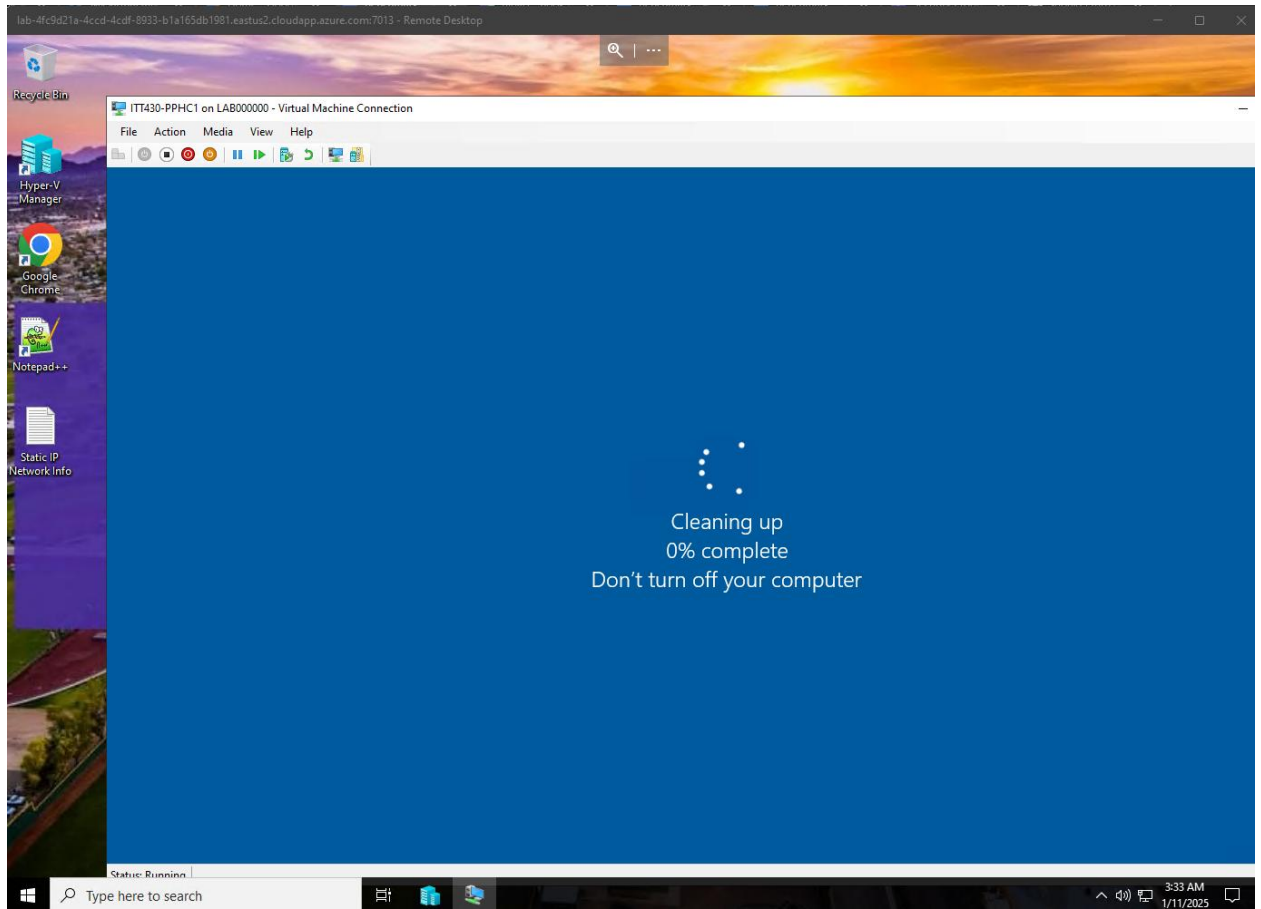


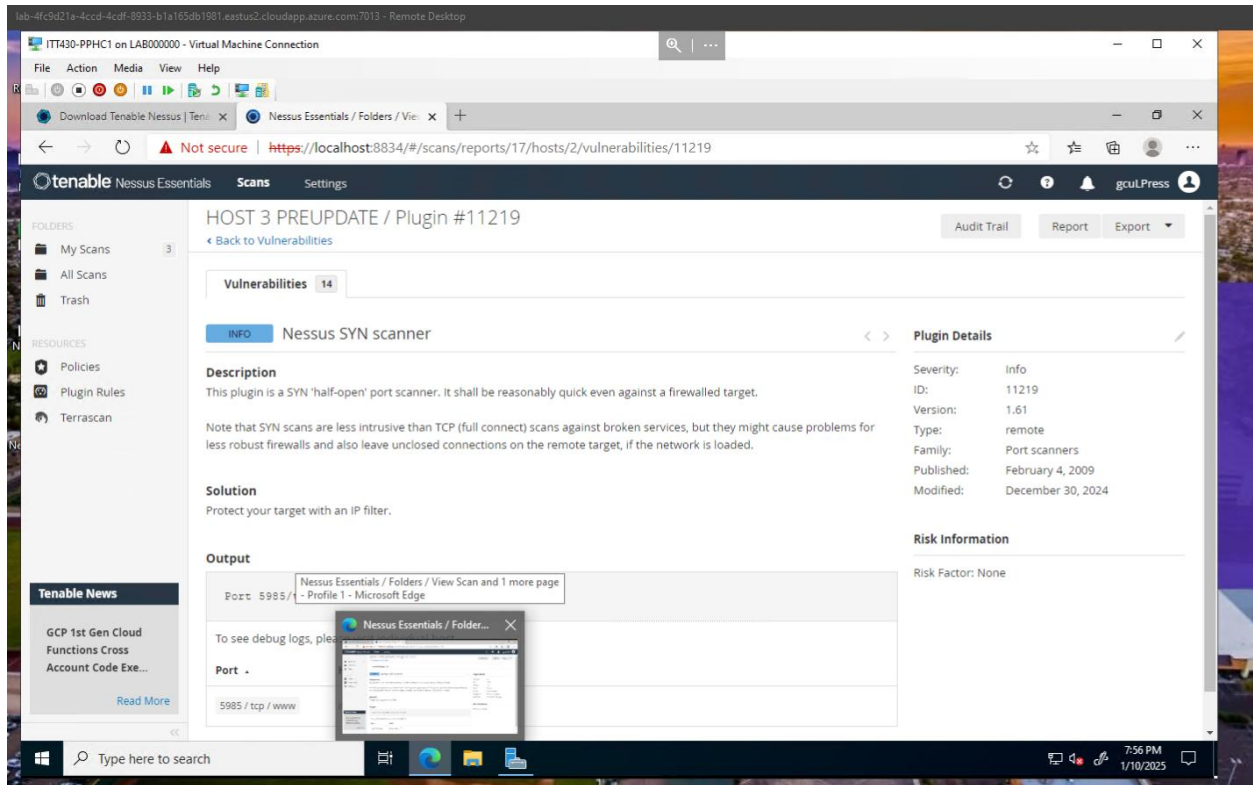
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References

Gracy, M. (2024, December 12). *12 types of network security measures*. Sprinto.

<https://sprinto.com/blog/network-security-measures/>